

Q2: How many solutions?

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$$x_1, x_2 = \frac{-b \pm \sqrt{\Delta}}{2a}$$

Problem: Can's take $\sqrt{}$ when $\Delta < 0$!

Solution: Complex numbers $\mathbb{C}$

$$i^2 = -1$$

Answer: Always 2 solutions over $\mathbb{C}$ counted with multiplicity!

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Fundamental Theorem of Algebra

*A degree d polynomial always has d roots
when counted with multiplicities over $\mathbb{C}$*